PHYSICAL AND CHEMICAL SCIENCE

Heating, Cooling and Dissolving

Melting, freezing, cooking and boiling, changes that can and cannot go back, and dissolving



ice water hot water chocolate

In this pack

Learn it: Heating, cooling and dissolving ★☆☆
Fact check: Change facts ★☆☆
Think it through: Can it change back? ★★☆
Investigate: Where does ice melt fastest? ★★☆
Real-world case study: Making pitha in winter ★★★
Review and challenge: Show what you know ★★★

This pack belongs to: _____ **Class:** _____

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LEARN IT · C2 · HEATING AND COOLING

Heating, cooling and dissolving

★☆☆ Grade 1

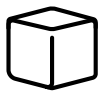
Name: Class: Date:

Learning goals

- Describe how heating and cooling change things.
- Find things that dissolve.

Some materials **change** when we heat them or cool them.

When we heat it...	When we cool it...
ice **melts** into water	water **freezes** into ice
chocolate and butter **melt**	melted chocolate goes **hard** again
an egg **cooks** and goes hard	a cooked egg stays cooked
water **boils** and turns to steam	steam turns back into water drops



ice



chocolate



egg



hot water

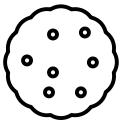
Some changes can go **back** (ice ↔ water). Some changes **cannot** go back (a cooked egg).

Dissolving

When we stir **sugar** or **salt** into water, it seems to disappear. It has **dissolved**. Sand and rice do not dissolve.



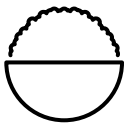
salt dissolves



sugar dissolves



sand does not



rice does not

⚠ Stay safe

Only grown-ups use the stove and hot water.

Key words

melt change from solid to liquid when heated

freeze change from liquid to solid when cooled

dissolve mix into water and seem to disappear

FACT CHECK · C2 · HEATING AND COOLING

Change facts

★☆☆
Grade 1

Name: Class: Date:

Learning goals

- Sort changes.
- Find what dissolves.

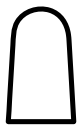
1 Heating or cooling?

SORT THESE

an egg cooks **ice melts** **water freezes** **melted chocolate goes hard**
water boils **jelly sets**

Heating	Cooling

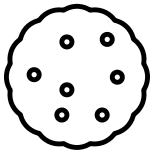
2 Circle the things that dissolve in water.



salt



sand



sugar



coin

THINK IT THROUGH · C2 · HEATING AND COOLING

Can it change back?

★★☆
Grade 1

Name: Class: Date:

Learning goals

- Explain reversible changes.

1 Can an ice lolly that melted be frozen again?

◯ yes ◯ no

2 Can a cooked egg become raw again?

◯ yes ◯ no

3 Why does ice cream melt on a hot day?

INVESTIGATE · C2 · HEATING AND COOLING

Where does ice melt fastest?

★★☆ Grade 1

Name: Class: Date:

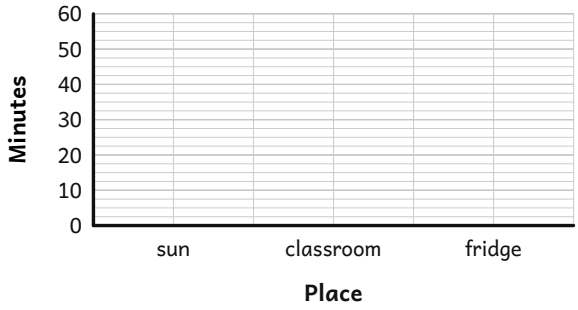
Learning goals

- Do a fair test.
- Read a bar chart.

Class 1 put the same size ice cubes in three places and timed how long they took to melt.

1 Our results. Draw the bars.

Place	in the sun	in the classroom	in the fridge
Minutes to melt	10	25	60



2 Where did the ice melt fastest? Why?

REAL-WORLD CASE STUDY · C2 · HEATING AND COOLING

Making pitha in winter

★★★
Grade 1

Name: Class: Date:

Learning goals

- Read about heating food.

In winter, Grandma makes **pitha**. She mixes rice flour with water to make a soft dough. She melts **date palm gur** (jaggery) in a pan.

She steams the pitha over hot water until it is cooked. The soft dough changes into a firm cake. Yum! The changes from heating cannot go back.

1 **What does heating do to the gur?**

.....

2 **Can the cooked pitha turn back into dough?**

◯ yes ◯ no

REVIEW AND CHALLENGE · C2 · HEATING AND COOLING

Show what you know

★★★
Grade 1

Name: Class: Date:

Learning goals

- Use change words.

1 Find the words.

Y	I	C	E	H	A	M
M	V	P	D	E	I	Y
G	O	R	S	A	L	T
T	M	V	D	T	W	B
I	E	O	C	O	O	L
L	L	T	W	P	A	B
N	T	C	P	J	G	L

cool heat ice melt salt

2 Water turns into ice when it is...

◯ heated ◯ cooled

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
say what heating and cooling do	◯	◯	◯
tell if a change can go back	◯	◯	◯
find things that dissolve	◯	◯	◯